OMAR KHATAB

Frontend Developer — React / Next.js / TypeScript

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SUMMARY

Frontend Developer (React / Next.js) with a Mechanical Power Engineering background. I combine engineering problem-solving with modern frontend to build practical, well-tested tools — e.g., an HVAC calculator that automates BTU cooling load & duct-sizing formulas with a transparent 'show your work' breakdown and unit-tested calculation logic. Completed military service, based in Cairo, open to remote / hybrid roles.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), HTML5, CSS3
- **Frontend Core:** React.js, Next.js (App Router, Server Components), Context API, REST API
- **Styling:** Tailwind CSS, Material UI (MUI), Framer Motion, Responsive Design (320px–1920px)
- **Testing:** Vitest (unit tests for calculation logic)
- **E-Commerce:** Shopify (Liquid, Ajax Cart API, Section Rendering API), Shopify CLI
- **Tools:** Git, GitHub, Vercel, Windsurf (AI-assisted dev)

PROJECTS — PRACTICAL EXPERIENCE

Shopify Theme Development — Custom E-Commerce Sections Code | *Sep 2026*

- Translated Figma to custom Dawn theme sections (Banner + Product Grid with Quick View) using Liquid, vanilla JS, Ajax Cart & Section Rendering APIs.
- Implemented O(1) hash-map variant selector, Black+Medium cross-sell auto-add, and no-reload header cart refresh. Handled race conditions by locking the popup during async requests. Used a feature-branch + PR workflow.

HVAC Duct & BTU Calculator Live Demo | Code | *Sep 2026*

- Built a multi-page engineering tool with Next.js App Router & TypeScript that estimates BTU cooling load and sizes rectangular ducts, with a transparent formula breakdown.
- Extracted calculations as pure, testable functions and created a reusable typed input component (slider + validation) shared across calculators.
- Added Vitest unit tests covering the core calculation functions to guard against regressions.

E-Commerce Store — React Live Demo | Code | *Jul–Aug 2026*

- Built a full-featured storefront with React, React Router, and Context API — including live product data via a REST API (DummyJSON), error handling, and skeleton loading states.
- Persisted cart/favorites with localStorage, added search and category filtering, and used Framer Motion for a mobile-first responsive UI.

Earlier Work — Responsive Pages (HTML5, CSS3, Vanilla JS) Live Demo | Code | *2021–2024*

- Built 5 responsive pages + an admin dashboard from PSD/Figma before moving to React/Next.js: semantic HTML5, BEM, Flexbox/Grid, vanilla JS — no frameworks.

ADDITIONAL BACKGROUND — MECHANICAL POWER ENGINEERING

- **Internships (Aug–Sep 2022):** National Authority for Tunnels & Arab Contractors — exposure to metro HVAC, ventilation & fire-fighting systems; engine assembly.
- **Engineering Projects:** Arduino-based PID Temperature Control (2023–2024); Solar Cell Cooling Graduation Project — tested designs lowering panel temperature by 10–20°C; Wind Turbine Blade-Pitch controller model (2021–2022).

LEARNING & EDUCATION

- **Currently Learning:** Advanced React.js (Comprehensive Course, Académie Tarmiz) — In Progress
- **BSc, Mechanical Power Engineering** — Ain Shams University | Cairo, Egypt | 2019–2024 | Military Service: Completed
- **Languages:** Arabic (Native), English (Professional Working Proficiency — technical reading, writing, documentation)